15–SHIP RESISTANCE AND POWERING

15.0 INTRODUCTION

Ship propulsion is the process of moving a vessel from one point to the other at a desired speed. This may be achieved:

- Naturally by using wind energy/sails, or
- Artificially using:
  - Manual power/paddles, or
  - Mechanical power

To achieve this successfully, the resistance to motion must first be determined and overcome.

15.1 SHIP RESISTANCE

It is known that:

$$\text{Power (P)} = \text{Force(F)} \times \text{Velocity(V)}$$

For a ship, this force equals the ***Total Resistance (R_T)*** that must be overcome to drive it at speed ***(V)***.

If a ship was therefore to be towed at constant speed in calm weather without any interference effects from the towing ship, R_T will be the force or tension in the rope. Therefore, R_T is sometimes referred to as the Tow-Rope Resistance. Figure 15.1.

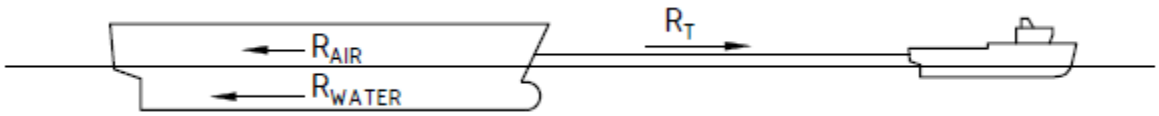


Figure 15.1 – Total or Tow-Rope Resistance

15.2 COMPONENTS OF SHIP RESISTANCE (Ref:

15.2.1 – Calm Weather Resistance

In calm weather, i.e. no wind and still water conditions, R_T is generated entirely by the ship's hull and is made up of the following:

1. Water resistance (R_{WATER})
2. Air resistance (R_{AIR})

Figures 15.2 to 15.4 are diagrammatic representations of the components of ship resistance.

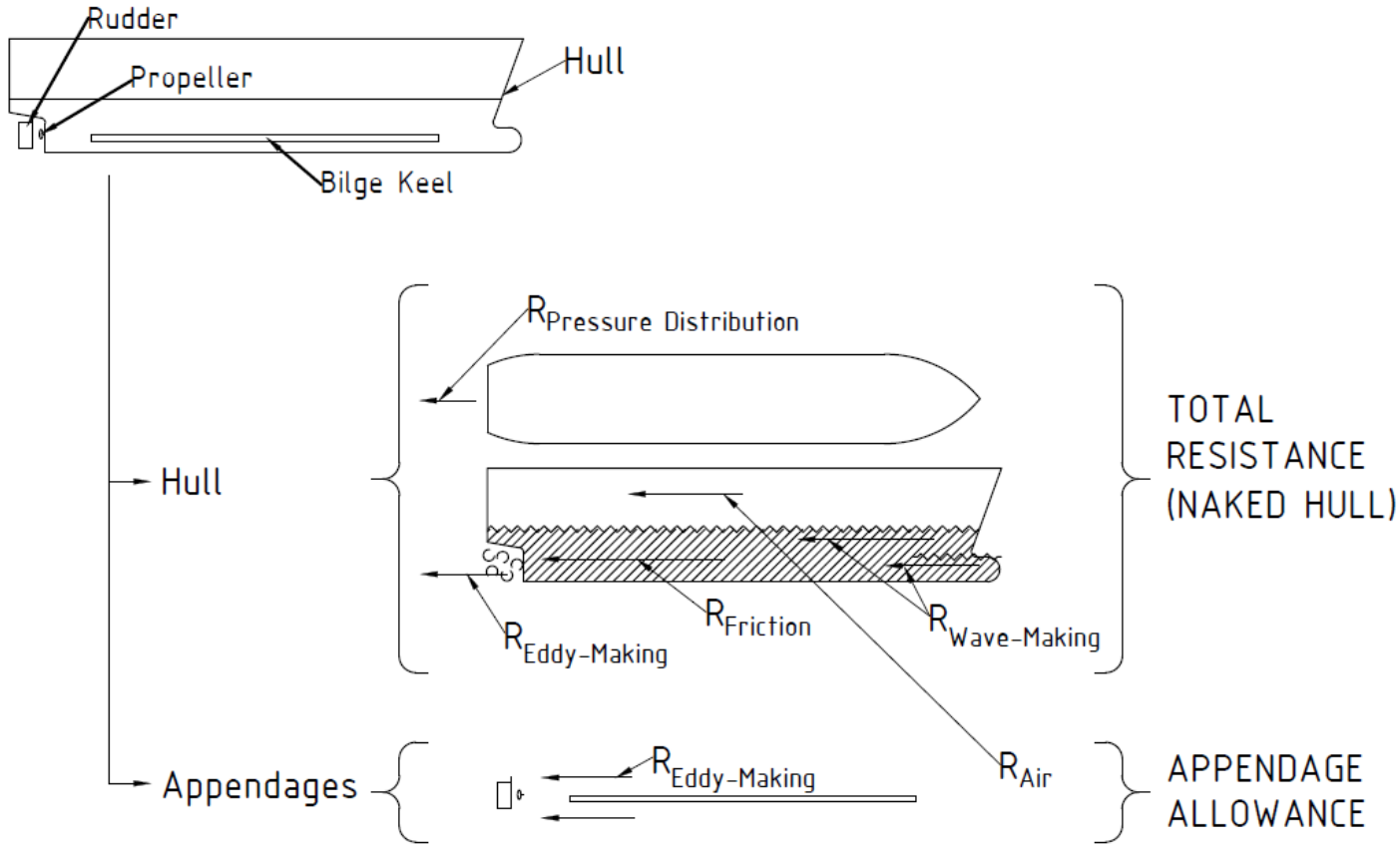


Figure 15.2 – Components of Ship Resistance

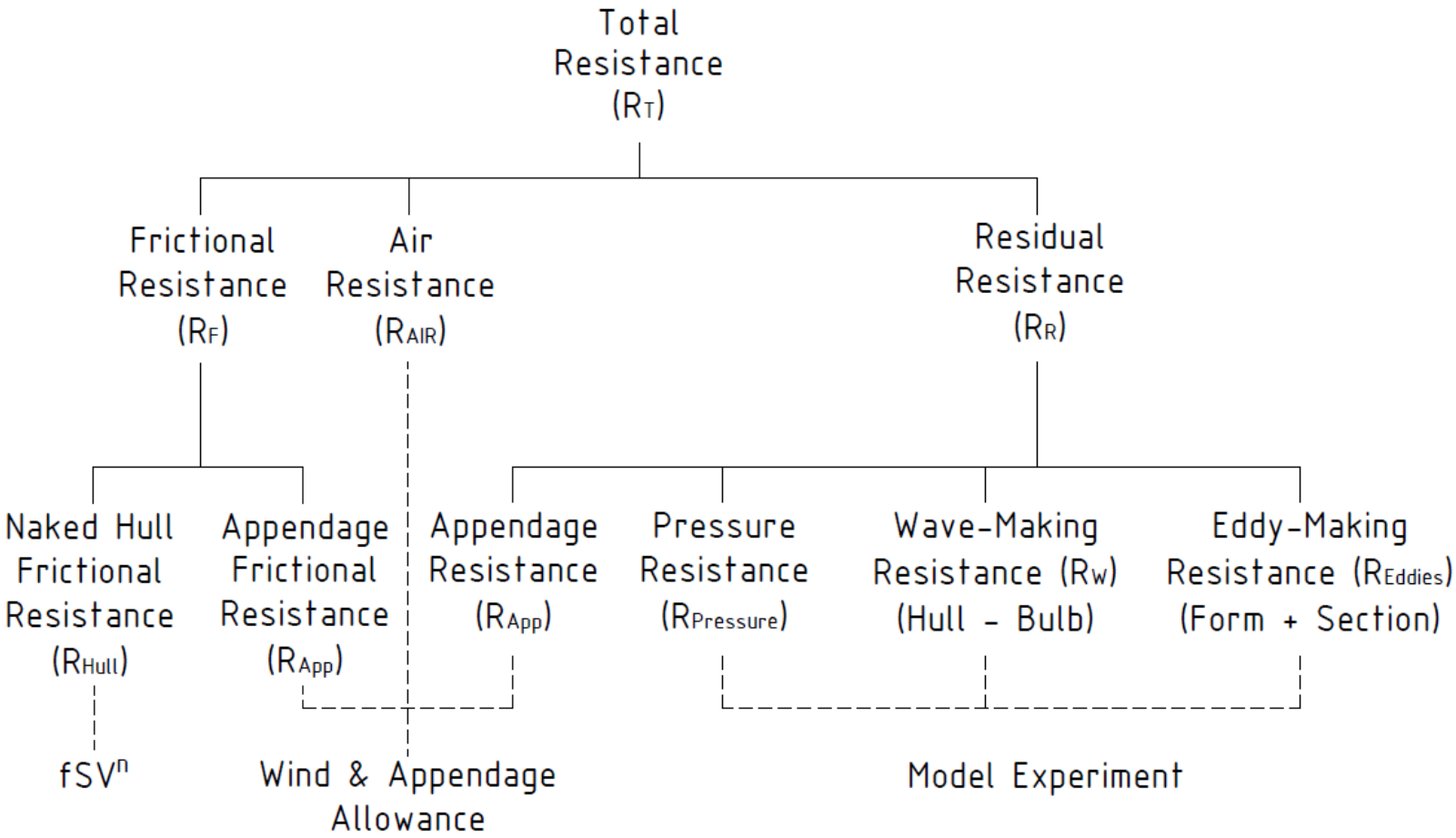


Figure 15.3 – Components of Total Resistance

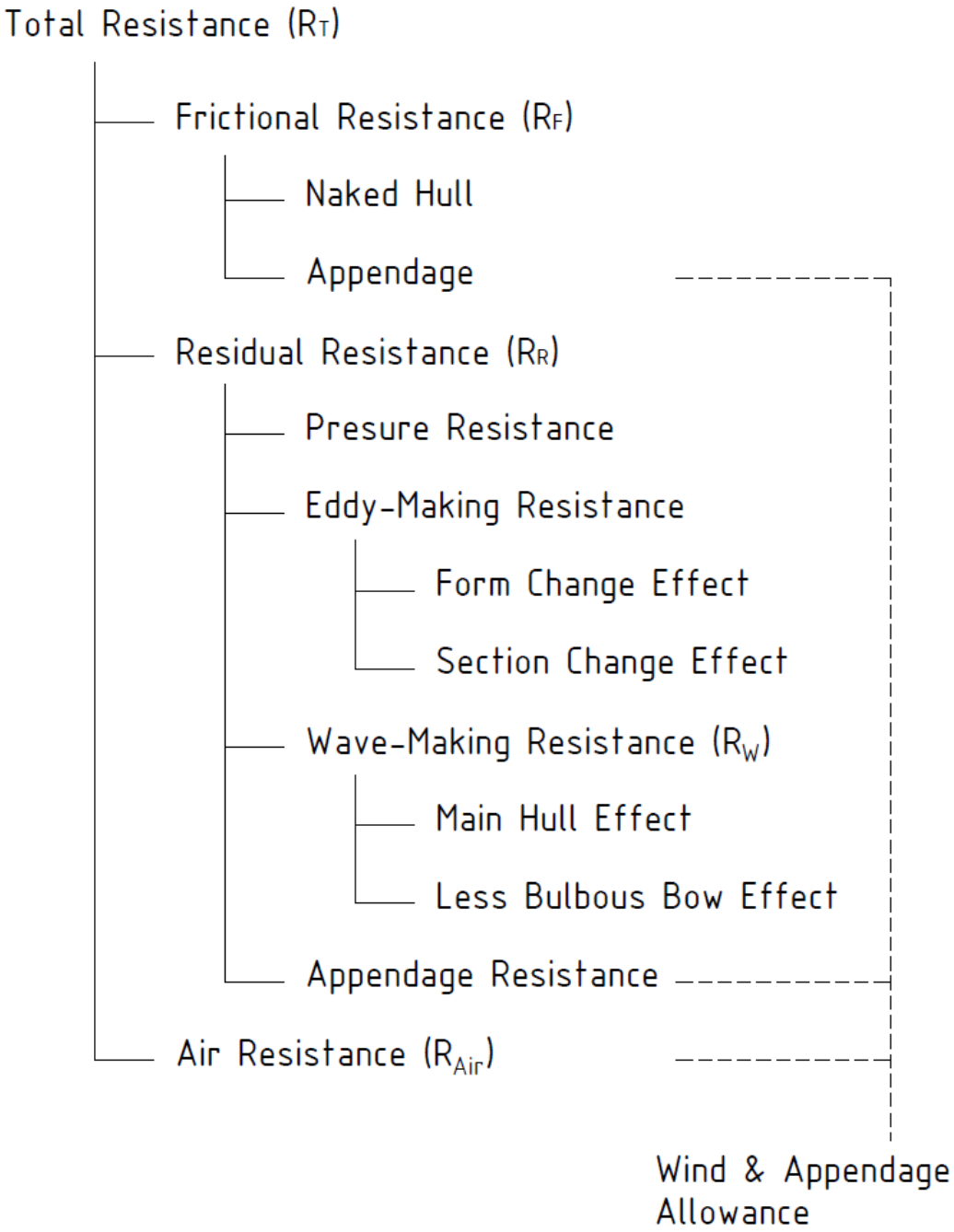


Figure 15.4 – Resistance Tree

15.2.2 – Components of Ship Resistance Explained

15.2.2.1 - **Water Resistance** (R_{WATER}):

Naked hull resistance

1. Frictional Resistance (R_F)

Due to the property of viscosity the thin layer of water directly in contact with the hull is dragged along as it attempts to shear off a ship which is moving forward at speed (V). This layer in turn drags the next layer further out and so on with rapidly decreasing energy and shearing force until at some distance measured perpendicularly away from the hull's surface, velocity reduces to zero.

This is evidenced by a divergent body of water referred to as the Boundary Layer. Its thickness at any section is defined by the point at which the forward velocity (v) has reduced to 1% of ship speed. The width is comparatively small and close to the hull at the bow because of variation in the forward speed of the water along the length which is rapid and more pronounced up front. See Figure 15.5.

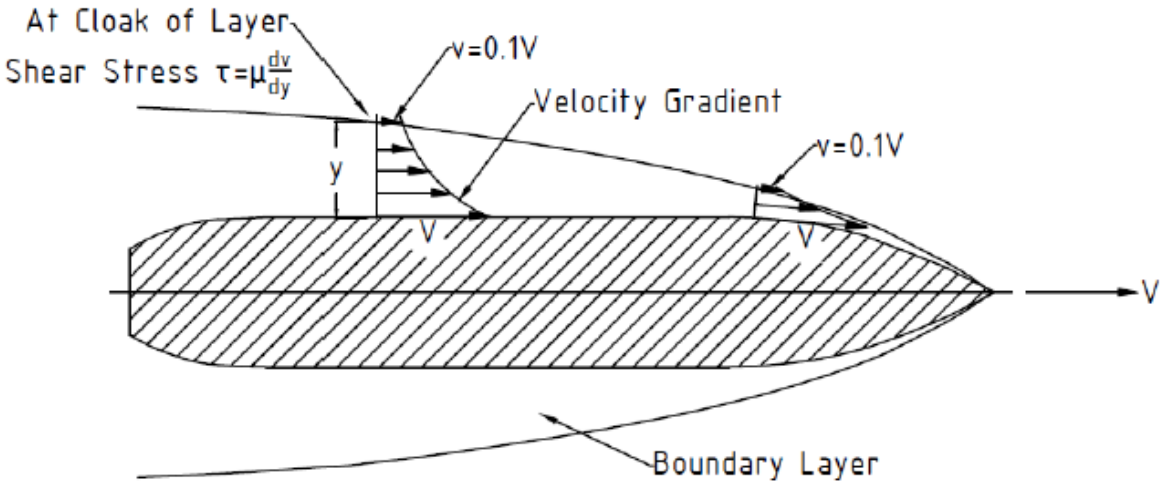


Figure 15.5 – Boundary Layer & Naked Hull Friction

The net effect of this drag is hence referred to as ***Viscous Resistance*** or ***Frictional Resistance***. Frictional resistance is influenced by flow pattern and increases with turbulence as shown by the Drag Coefficient/Reynolds Number curves in Figure 15.6. Where:

Drag Coefficient: ($C_D = D/(\rho A V^2/2)$, and
Reynolds Number: ($Re = Vl/v$)

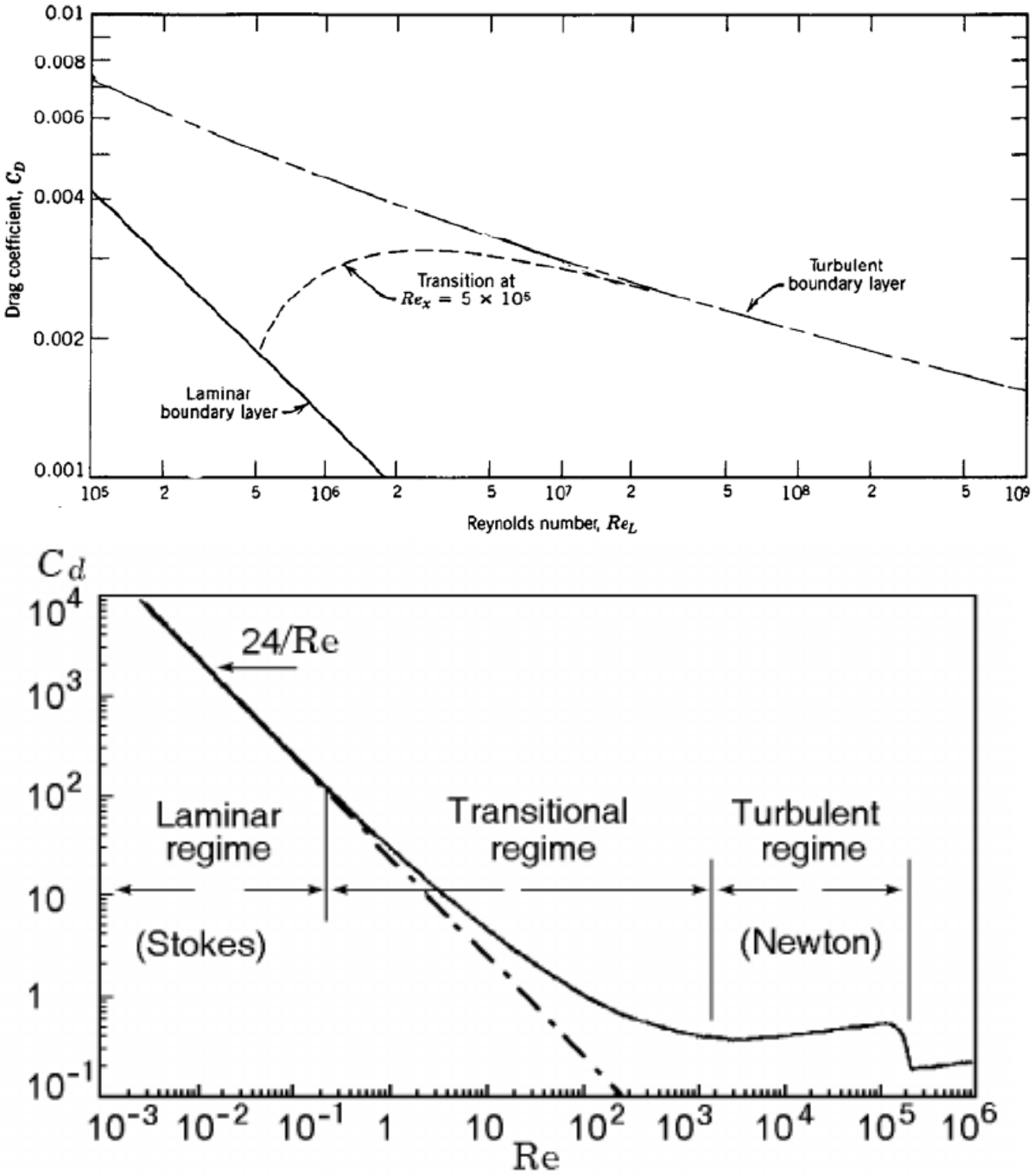


Figure 15.6 – Flow Pattern/Drag Coefficient Curves

2. Residual Resistance (R_R)

- Pressure Drag Component

  A ship at rest in still water experiences hydrostatic pressures which act normal to the immersed surface. When resolved transversely and longitudinally the resultants will be zero in either direction. See Figure 15.7.

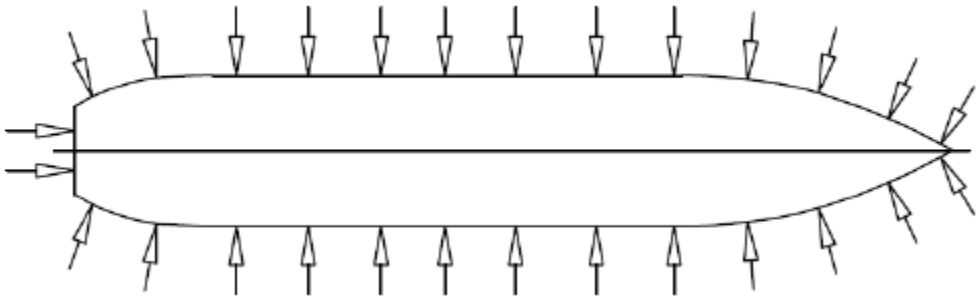


Figure 15.7 – Pressure Distribution at Rest (No Flow)

When now a ship is moved forward, dynamic pressures are generated which when resolved will be zero transversely, while longitudinally pressures will increase up front and reduce aft to cause flow aft and hence a drag as in Figure 15.8.

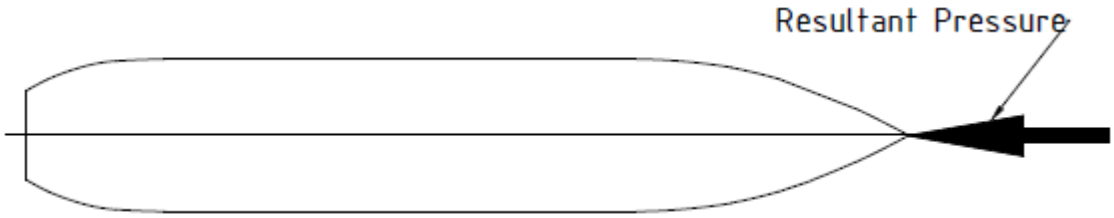


Figure 15.8 – Pressure Distribution when in Motion (Net Drag)

- Wave-making component (R_W).

Main Hull

When a point moves along the surface of water the combined effects of pressure, gravity and surface tension creates a disturbance and results in the formation of two distinct sets of waves namely:

  - Divergent waves - with a series of diagonal or oblique crests moving outwardly, and
  - Transverse waves - which appear as troughs and crests along the length.

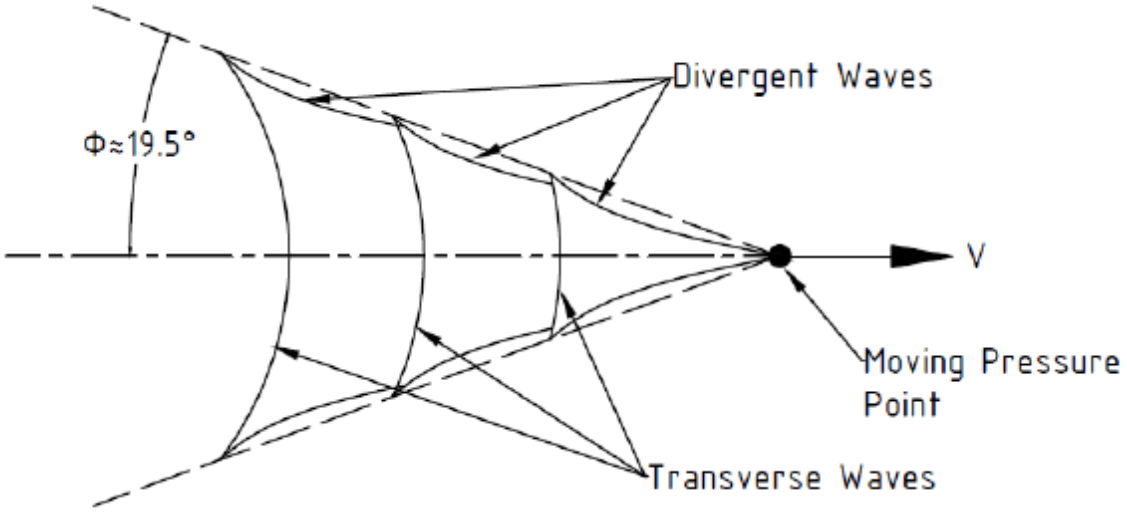


Figure 15.7 – Wave Pattern of a Single Moving Point

When this phenomenon is applied to a ship's hull the equivalent of two moving points come into play and results in two sets of waves as shown in Figure 15.8. These are:

- A bow wave system at the forward, and
- A stern wave system at the aft

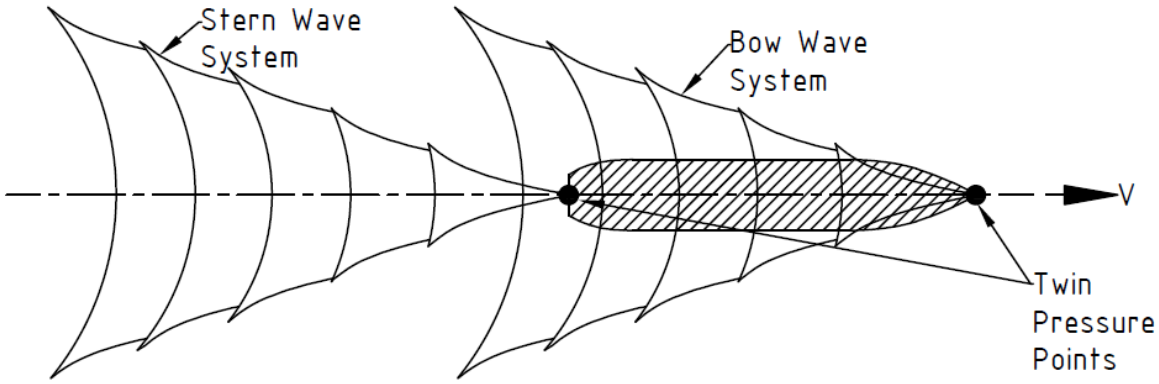


Figure 15.8 – Wave Pattern of Two Moving Points

A bow wave creates a crest at about a quarter the wavelength (I.e. $\frac{\lambda}{4}$) from the pressure point while a stern wave creates a trough. Now:

$$\lambda = \frac{2\pi}{g} V^2$$

Therefore, the length between crests will increase with speed.

Divergent waves do not interfere much with each other. Transverse waves on the other hand greatly do and as speed increases it is possible for a crest

from the bow system to coincide with a crest from the stern thereby resulting in reinforcement. If in the other hand a crest coincides with a trough, then cancellation will result. See Figure 15.9.

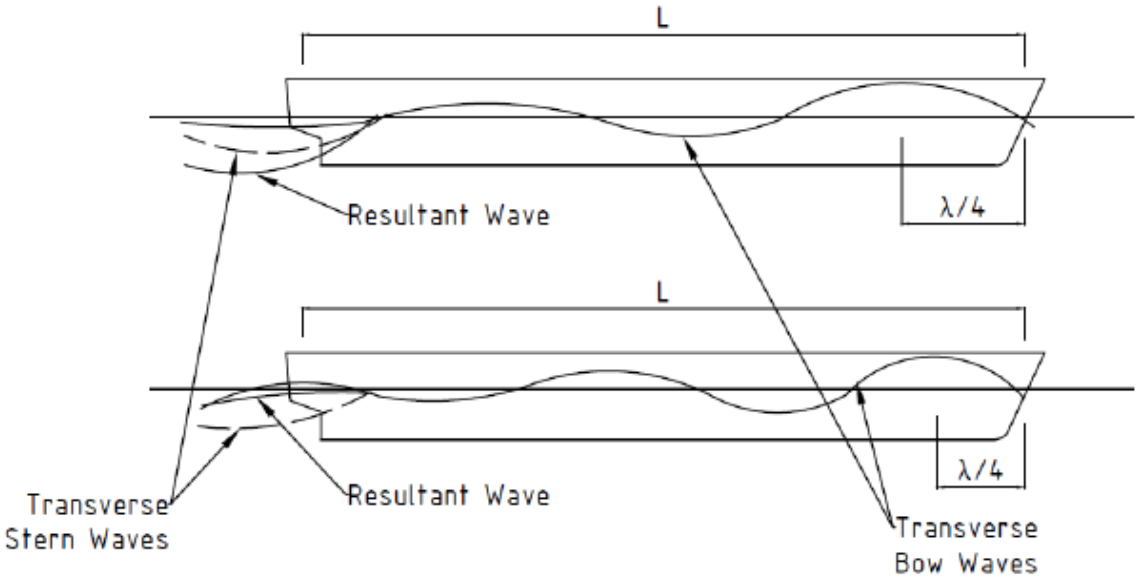


Figure 15.9 – Stern Wave Reinforcement or Cancellation

Bulbous Bow Effect

This is a special bulb of approximately spherical construction that is attached to a normal stem just below water level to modify wave-making characteristics forward. Acceleration of the bulb under the surface induces a low pressure field which extends towards the surface. This low pressure field then interacts to cancel out or reduce the bow pressure wave as shown in Figure 15.9. The overall effect is therefore a reduction in energy absorbed to make waves.

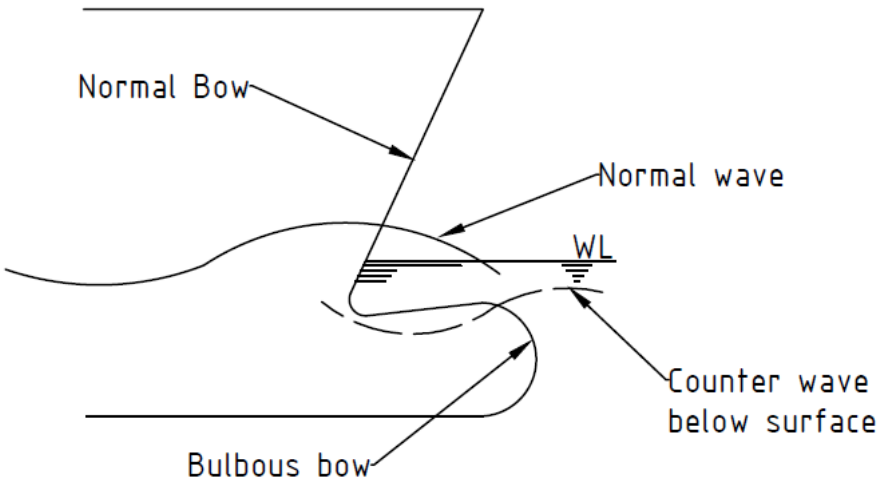


Figure 15.9 – Effect of Bulbous Bow

- Eddy-making component (R_{EDDY})

Section Change Effect

Whenever there is a sudden change in section such as an immersed transom stern construction or turn of bilge, abrupt obstructions to flow will occur causing flow separation and formation of vortices or eddies (i.e. movement in a stream of fluid in which a section of current doubles back on itself causing a miniature whirlpool to oppose the main current). See Figure 15.10. This results in pressure loss behind a ship and a net drag on the hull which can be considered as resistance to motion.

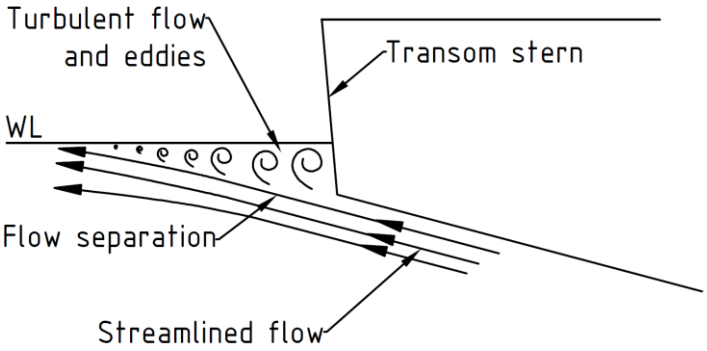


Figure 15.10 – Eddy-Making Resistance due to Section Change

Form Change Effect

Due to viscous effects and the sudden change in hull form aft, water leaving at the rear of a ship separates and forms eddies. See Figure 15.11. The result is that pressure is not fully recovered aft to balance the pressure up front and as consequence resistance is generated or there will be viscous drag due to form change

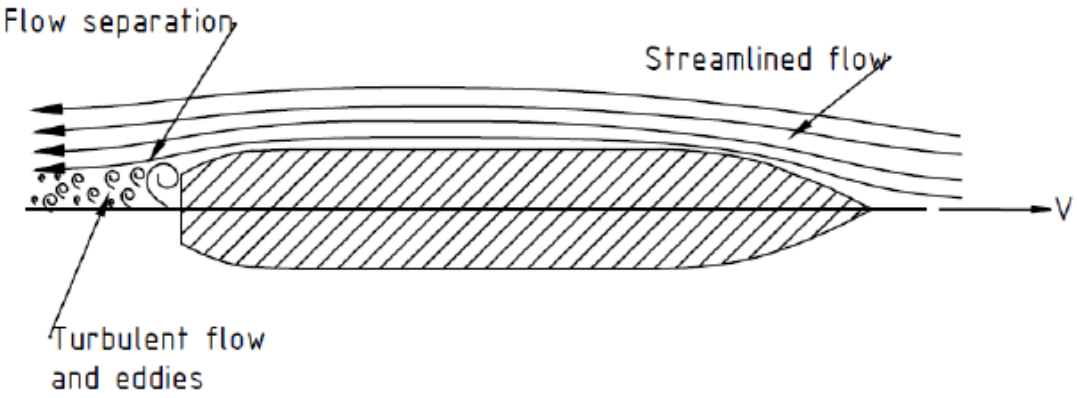


Figure 15.11 – Eddy-Making Resistance due to Form Change

Appendage Resistance

Appendages comprise the following:

- Rudder,
- Bilge keels,

- Stabilizers,
- Transverse thruster openings,
- Propeller and bossing,
- Shaft brackets, etc.

Like the naked hull, these appendages individually generate frictional and wave-making resistances that must be accounted for.

WAKE
The net effect of generated waves, and drag due to friction is a body of water which is dragged along or follows the ship for some considerable distance as it moves. This body or mass of water is called the ***Wake*** and has a net forward velocity $\boldsymbol{V_W}$ referred to as ***Wake Speed***.

15.2.2.2 - **Air Resistance** (R_{AIR}):
Air being a fluid will impose frictional drag on the exposed surface of a hull just like water. In still air or calm weather conditions, air resistance is considered to be minimal and hence supplemented for by including its effect in the appendage allowance.

15.2.3 – Rough Weather Resistance
In rough weather R_T is made up of:

- Calm weather resistance, plus
- A variety of resistances from sources such as:
    - Strong winds
    - Non-ship generated waves
    - Breaking waves, etc.

SUMMARY
The total resistance (R_T) in calm weather is therefore made up of the following components.

1. Frictional resistance. Caused by the motion of the hull through a viscous fluid (R_F).
2. Residuary resistance (R_R) which comprises:
    a. Pressure resistance. Drag aft resulting from high pressures generated forward and depressions generated aft that is caused by the motion of a vessel.
    b. Wave making resistance. Equivalent to the energy that must be supplied continuously by the ship to the wave system generated on the free surface. Includes:
        i. Naked hull component.
        ii. Bulbous bow component.
    c. Eddy resistance. Equivalent to the energy carried away by eddies shed from the hull or appendages. This is especially severe at the stern where the water may be unable to follow the curvature and will break away from the hull. Includes:
        i. Form change component.

ii. Section change component.

3. Appendage resistance.
4. Air resistance imposed on superstructures and the above–water portion of the hull.

In addition, there are a variety of resistances such as resistance due to turning, drag caused by sailing in waters with restricted depth or restricted width at the sides of a ship. These are however not taken into consideration in design calculations.

Resistances 2 and 3 are usually considered together (although they arise from deferent physical mechanisms) under the name ***Residual Resistance*** (R_R).

Resistances under 4 do not scale in the same proportions from model to full size, i.e. when fitted onto a ship, as a naked model hull will scale to a full-size naked hull of a ship. Together with air resistance they are therefore supplemented for by adding 10 to 30% to the R_T after calculation, depending on the expected sailing routes.

Rough weather resistances are not usually factored into design calculations. As a result, whenever a ship encounters such conditions, especially a head wind, she is compelled to slow down.

15.3 DETERMINATION OF R_T IN CALM WEATHER

In order to avoid costly mistakes, it is necessary to estimate the power required for the desired speed long before a ship is built. This is achieved by first determining the R_T as follows:

15.3.1 – Frictional Resistance (R_F)

William Froude derived an empirical formula for calculating Frictional resistance from:

$$R_F = fSV^n$$

Where:

f = A coefficient that is dependent on:

- Length of surface (i.e. $f \propto \frac{1}{L}$)
- Roughness of surface (i.e. $f \propto Roughness$)
- Density (ρ) of water

For steel ships in sea water:

$$f = 0.417 + \frac{0.773}{L + 2.862}$$

S = Wetted surface area in m^2
V = Ship speed in knots, and
n = Coefficient which has a value 1.825 for normal ship forms

15.3.2 – Residual Resistance (R_R)

In the nineteenth century William Froude recognized through experiments that when a geometrically similar model of a ship is run, a similar wave pattern is created. The speed at which the model is run must however be lower than that of the ship to obtain this pattern.

By further studies to compare the specific resistance curves (R/Δ) of ships and models he also noted that they always exhibited a similarity of the form shown in Figure 15.12, with the curve for models always having greater values than that for ships.

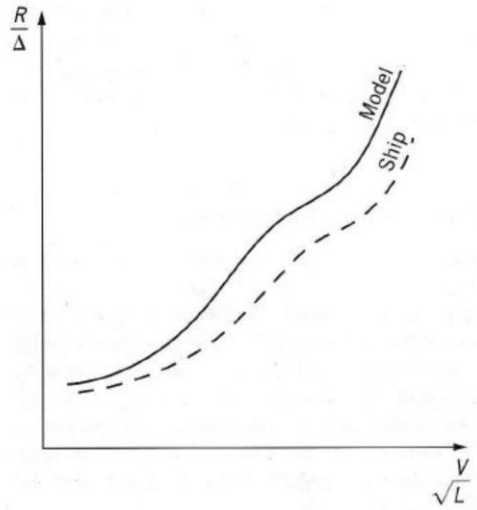


Figure 15.12 – Specific Resistance Curves

From this he concluded that:

"***The residuary resistances of geometrically similar ships are in the ratio of their displacements when their speeds are in the ratio of the square roots of their lengths***". This is derived from Froude's Number ($V/\sqrt{gl}$)

I.e. $R_R \propto \Delta$

When $V \propto \sqrt{L}$

This is referred to as the "Froude's Law of Similarity" and the speeds are referred to as the ***Corresponding Speeds***. On the basis of this law the total ship resistance is evaluated by using a model and thereafter co-relating the values so obtained to determine that for the intended ship. This is referred to as the Model Testing Method and is carried out as follows:

1. Tow a geometrically similar model of a ship (without appendages) at the corresponding speed V_M and measure the total resistance R_{TM} usually in fresh water.
2. Calculate the frictional resistance R_{FM} of the model based on coefficient (f) for fresh water.

3. Derive the residuary resistance R_{RM} of model by deduction to give:

$$R_{RM} = R_{TM} - R_{FM}$$

4. Correct the residual resistance of model for density:

$$R_{RM} = \frac{\rho_S}{\rho_M} \times R_{Rm}$$

5. Obtain the residuary resistance of ship:

$$R_{RS} = \frac{\Delta_S}{\Delta_M} \times R_{RM}$$

Now since:

$$\Delta \propto L^3$$

The residuary resistance may also be calculated from:

$$R_{RS} = \left(\frac{L_S}{L_M}\right)^3 \times R_{RM}$$

6. Calculate the frictional resistance of the full-size ship R_{FS}.
7. Finally obtain the total resistance of naked hull from:

$$R_T = R_{FS} + R_{RS}$$

15.4 POWERING

15.4.1 **Effective Power (P_E)**

Once total resistance R_T of a ship has been determined the power to drive it is then calculated. But then since this power has not taken into consideration the effects of appendages, it is referred to as ***Effective Power Naked (P_{EN})***. i.e.:

$$P_{EN} = R_T \times V$$

Finally, by adding the ***Appendage Allowance (Al)*** or the ***Ship Correlation Factor (SCF)*** as it is sometimes called, the ***Effective Power (P_E)*** becomes:

$$P_E = P_{EN} \times Al$$

15.4.2 **Thrust Power (P_T)**

Due to the motion of a ship a body of water is constantly dragged along for some distance until all the dragging energy is lost. This body of water is called the ***Wake*** and has a forward velocity ***(V_W)***. Now since the propeller travels within the wake then its relative velocity will be less than that of the ship speed by an amount equal to wake speed. This relative speed is referred to as the ***Speed of Advance (V_a)***. Therefore:

$$V_a = V - V_W$$

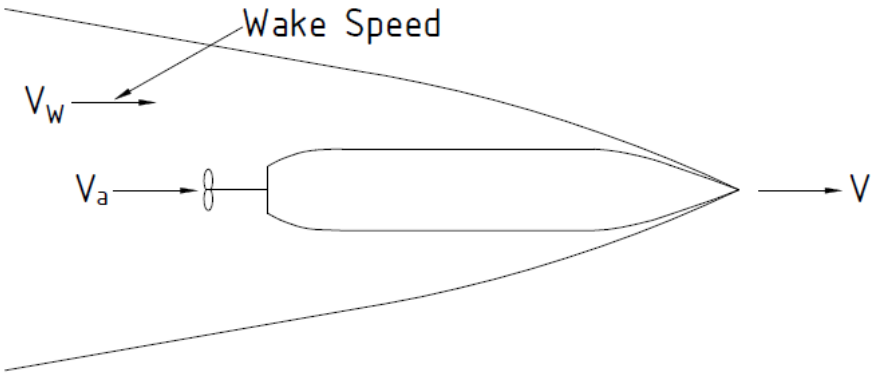


Figure 15.13 - Effect of the Wake Field

Hence for a thrust **(T)**, Thrust Power is equal to:

$$P_E = T \times V_a$$

In non-dimensional terms the effect of the wake is expressed as a fraction of ship speed by ***Taylor's Wake Fraction (w_t)***. Where:

$$w_t = \frac{Wake\ Speed}{Ship\ Speed} = \frac{V_W}{V} = \frac{V - V_a}{V} = 1 - \left(\frac{V_a}{V}\right), \text{ or}$$

$$V_a = V(1 - w_t)$$

Therefore:

$$P_T = T \times V(1 - w_t)$$

Now in principle (T) must be equal to R_T, but then acting like a pump, reduction in pressure occurs as water is accelerated into the propeller. This results in a suction effect on the stern of the hull which can be quantified as resistance to motion. This therefore requires additional thrust to be generated to overcome resistance if ship speed is to be achieved. Actual thrust generated is hence greater than R_T to cater for the shortfall.

$$T > R_T$$

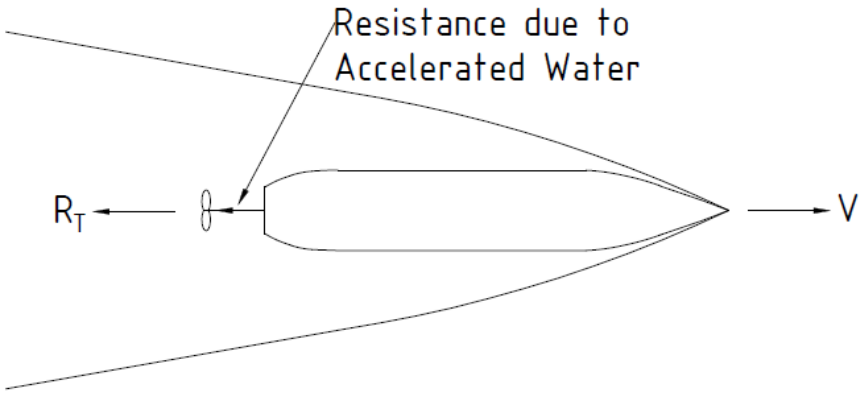


Figure 15.14 - Effect of Propeller Action

Or:

$$T = R_T + Augmenting\ Factor$$

Froude quantified this addition or augment as a fraction of the thrust and referred to it as the ***Thrust Deduction Factor (t)*** to give:

$$T(1 - t) = R_T$$

From which:

$$T = \frac{R_T}{(1 - t)}$$

And hence:

$$P_T = \frac{R_T}{(1 - t)} \times V(1 - w_t)$$

$$P_T = R_T \times V \times \frac{(1 - w_t)}{(1 - t)}$$

$$P_T = P_E \times \frac{(1 - w_t)}{(1 - t)}$$

$$\frac{P_E}{P_T} = \frac{(1-t)}{(1-w_t)}$$

The ratio:

$$\frac{(1-t)}{(1-w_t)} = \text{Hull Efficiency } (\eta_H)$$

Hull efficiency is usually about 98%.

15.4.3 **Delivered Power (P_D)**

The power that must be delivered to a propeller to generate the required thrust is referred to as "***Delivered Power***" (***P_D***) and is related to P_T by:

$$\frac{P_T}{P_D} = \eta_P$$

Where (***η_P***) is the "***Efficiency of Propeller Behind***" a ship

Efficiency of Propeller Behind

It has been found that the efficiency at which a propeller absorbs torque when tested in open water is higher than when it is behind a ship.



Figure 15.15 - Propeller in Open Water



Figure 15.16 - Propeller Behind

This difference in performance is referred to as Relative Rotative Efficiency (RRE or η_{RR}), and hence:

$$\eta_{RR} = \frac{\text{Efficiency of Propeller Behind } (\eta_P)}{\text{Efficiency of Propeller in Open Water } (\eta_O)}$$

From which:

$$\eta_P = \eta_{RR} \times \eta_O$$

Propeller behind efficiency ranges between 60 to 75%.

15.4.4 **Brake Power (P_B)**

Power available at the output shaft of an engine is referred to as the ***Shaft Power (P_S)***. It is measured by means of a power brake which applies a resistance torque as

brake on the shaft. For this reason, it is appropriately referred to as the ***Brake Power (P_B)***.

Transmission Efficiency

Now due to energy losses in gearboxes, shaft bearings and the stern tube, the power actually delivered to the propeller will be less than the brake power. Expressing these losses as ***Transmission Efficiency (η_T)***, then:

$$\frac{P_D}{P_B} = \eta_T$$

Transmission efficiency usually lies between 95 and 98%.

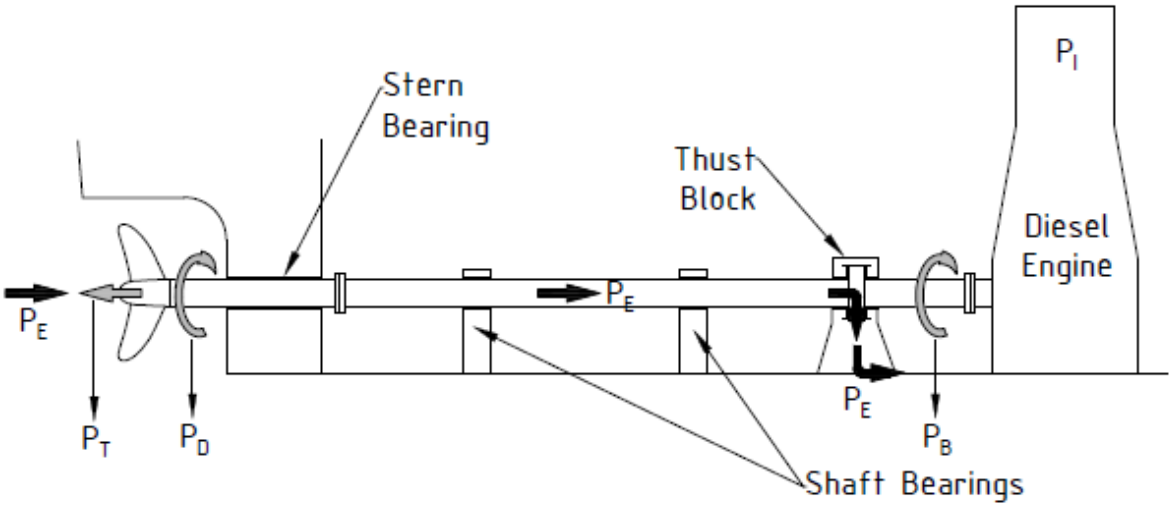


Figure 15.17 – RELATIONSHIP BETWEEN POWERS – DIESEL MACHINERY

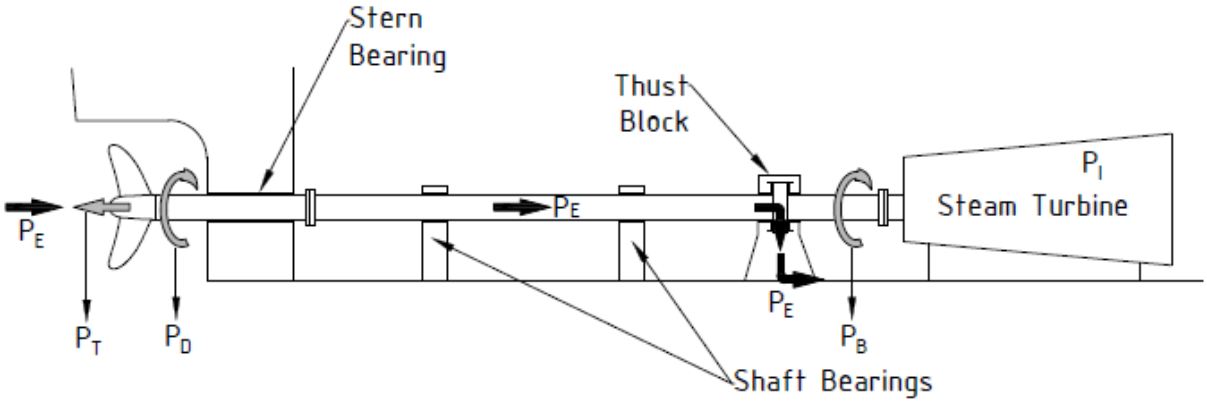


Figure 15.18 – RELATIONSHIP BETWEEN POWERS – STEAM TUNBINE MACHINERY

15.4.5 **Installed Power**

Over the life of a vessel it will grow in weight, due to added systems, ship alterations, and other various additions. Hull roughness will increase irrespective of regular dry-dock maintenance. The propelling machinery will deteriorate with age

and cannot continue to perform as new. These are changes that cannot be prevented no matter how and must therefore be provided for at the design stage. Additionally, in heavy seas resistance will increase as explained earlier. A ***Sea Margin*** of 15% power is therefore added to cater for increases in weight, hull roughness and rough weather, while an ***Engine Margin*** of 10% power is added for loss of machinery performance. The total installed power will therefore read:

$$\text{Installed Power} = (1 + 0.15 + 0.10) \times \text{P}_\text{B} = 1.25 \times \text{P}_\text{B}$$

15.4.6 Propulsion Efficiency

The ratio of effective power to shaft or brake power represents the overall efficiency of propulsion machinery and is called the ***Propulsive Coefficient (PC)***.

$$\text{PC} = \frac{\text{P}_\text{E}}{\text{P}_\text{B}} = \frac{\text{P}_\text{E}}{\text{P}_\text{S}} = \eta_\text{H} \times \eta_\text{P} \times \eta_\text{S}$$

When however, the comparison is made between effective power and delivered power, the propulsion efficiency is referred to as the ***Quasi-Propulsive Coefficient (QPC)***.

$$\text{QPC} = \frac{\text{P}_\text{E}}{\text{P}_\text{D}} = \frac{\text{P}_\text{E}}{\text{P}_\text{T}} \times \frac{\text{P}_\text{T}}{\text{P}_\text{D}} = \eta_\text{H} \times \eta_\text{P} = \eta_\text{H} \times \eta_\text{RR} \times \eta_\text{O}$$

15.4.7 Other Efficiencies

Mechanical Efficiency

Brake or shaft power is obtained by converting energy in fuel to work. The input power from this conversion is referred to as ***Indicated Power (P_I)****. During the process losses will definitely be registered. These losses are quantified as mechanical efficiency (η_M). Therefore:*

$$\frac{P_B}{P_I} = \eta_M \quad OR \quad \frac{P_S}{P_I} = \eta_M$$

Mechanical efficiency of diesel engines is usually between 88 and 93%

Thermal Efficiency

Comparing power expected from the ideal cycle upon which the diesel engine works with indicated power achieved in practice, there is a difference. This is due to losses from incomplete combustion, cooling, heat addition, gas exchange and exhaust processes. This is quantified as thermal efficiency η_{TH}. Naming the expected power ***Thermal Power (P_{TH})****, then:*

$$\frac{P_I}{P_{TH}} = \eta_{TH}$$

Theoretical Thermal Efficiency

If finally, the ideal cycle is compared to the Carnot cycle as the ultimate then it can be said that the ***Theoretical Thermal Power (P_{TT})*** *is related to thermal power by:*

$$\frac{P_{TH}}{P_{TT}} = \eta_{TT}$$

15.5 POWER ESTIMATION

In the earlier stages of the design of a ship it is helpful to obtain an approximation to the power of the ship without restoring to model experiments. One system which may be used is the admiralty coefficient method. This is based on the assumption that for small variations in speed the total resistance of ships may be assumed to be proportional to:

$$R_T \propto R_{FS} \propto fSV^n$$

Now:

$$f \propto \frac{\rho}{L}$$

Therefore

$$R_T \propto \rho SV^n$$

But then:

$$S \propto \Delta^{2/3}$$

Therefore, if density is considered constant then:

$$R_T \propto \Delta^{2/3} V^n$$

Multiplying both sides by V:

$$R_T V \propto \Delta^{2/3} V^{n+1}$$

I.e.:

$$P_E \propto \Delta^{2/3} V^{n+1}$$

But:

$$P_E \propto P_B$$

Therefore:

$$P_B \propto \Delta^{2/3} V^{n+1}$$

Most merchant ships may be classed as slow or medium speed, and for such vessels the index 'n' may be taken as 2. Therefore:

$$P_B \propto \Delta^{2/3} V^3$$

Or:

$$P_B \times \text{Constant} = \Delta^{2/3} V^3$$

This constant is known as the ***Admiralty Coefficient (A_C)***. Therefore:

$$A_C = \frac{\Delta^{2/3} V^3}{P_B}$$

For similar ships, the Admiralty coefficient is the same. Values range between about 350 and 600, the higher values indicating more efficient ships.

15.6 FUEL ESTIMATION

15.6.1 Fuel Coefficient & Daily Consumption

From the performance characteristics of a power plant it is observed that within a certain speed range the ***Specific Fuel Oil Consumption (SFOC)*** is practically constant as shown in Figure 15.20.

This is the range within which a power plant operates most efficiently. If a vessel is operated outside this region consumption will increase by a margin proportional to the difference between the best and actual SFOC curve.

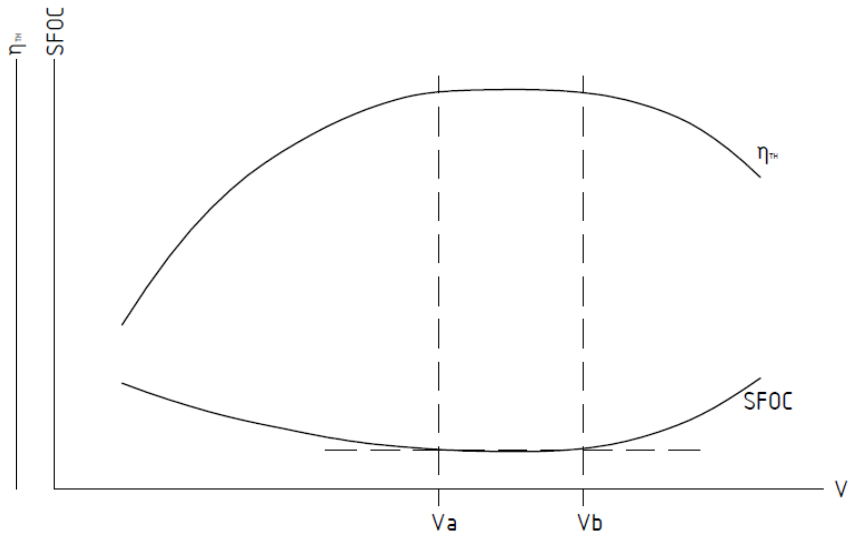


Figure 15.19 – PERFORMANCE CHARACTERISTICS

Where:

$$\text{SFOC} = \frac{\text{Fuel consumed in (kg)}}{\text{Power developed in an hour}}$$

Therefore, within the constant speed range:

$$\frac{\text{kg}}{\text{kWh}} = \text{Constant}$$

$$\frac{\text{kg}}{\text{h}} = \text{kW} \times \text{Constant}$$

Or:

$$\text{Fuel consumed/hour} \propto \text{Power developed}$$

From Admiralty Coefficient considerations:

$$P_B \propto \Delta^{2/3} V^3$$

Therefore:

$$\text{Fuel consumed/hour} \propto \Delta^{2/3}V^3$$

By extension fuel consumed in a day:

$$C \propto \Delta^{2/3}V^3$$

Or:

$$C \times \text{Constant} = \Delta^{2/3}V^3$$

This constant is referred to as the ***Fuel Coefficient (F_C)***. Therefore:

$$F_C = \frac{\Delta^{2/3}V^3}{C}$$

Its value is the same for similar ships.

15.6.2 Voyage Consumption
From Fuel Coefficient considerations, daily consumption:

$$C \propto \Delta^{2/3}V^3$$

Now voyage consumption (C_V) equals daily consumption multiplied by the number of days or time (T) spent. Therefore:

$$C_V \propto \Delta^{2/3}V^3T$$

And since time equals voyage distance divided by the speed (V), then consumption may also be expressed in the form:

$$C_V \propto \Delta^{2/3}V^2D$$

Reynolds Number is an important dimensionless quantity in fluid mechanics used to help predict flow patterns in different fluid flow situations. It is the ratio of inertial forces to viscous forces and is a convenient parameter for predicting if a flow condition will be laminar or turbulent.

$$R_e = \frac{Internal\ Forces}{viscous\ Forces} = \frac{\rho VL}{\mu} = \frac{VL}{v}$$

Where:

- ρ is the density of the fluid (SI units: kg/m^3)
- u is the velocity of the fluid with respect to the object (m/s)
- L is a characteristic linear dimension (m)
- μ is the dynamic viscosity of the fluid (Pa.s or N.s/m^2 or kg/m.s)
- v is the kinematic viscosity of the fluid (m^2/s).

With respect to laminar and turbulent flow regimes:

- Laminar flow occurs at low Reynolds numbers, where viscous forces are dominant, and is characterized by smooth, constant fluid motion;
- Turbulent flow occurs at high Reynolds numbers and is dominated by inertial forces, which tend to produce chaotic eddies, vortices and other flow instabilities.[7]

The flow is

- Laminar when *Re < 2300*
- Transient when *2300 < Re < 4000*
- Turbulent when *4000 < Re*

Drag Coefficient is a factor representing the drag acting on a body. It is measured by:

$$C_d = \frac{2F_d}{\rho u^2 A}$$

Where:

- F_d is the drag force, which is by definition the force component in the direction of the flow velocity
- ρ is the mass density of the fluid
- u is the flow speed of the object relative to the fluid
- A is the reference area